

The corresponding p-values are reported at the bottom of table 4. In the OLS regression, the test rejects the hypothesis that the coefficient on the test variable is zero. This means that the model estimated using the logarithmic specification is inappropriate. A similar result is found for the OLS estimated using $\ln(1 + T_{ij})$ as the dependent variable and NLS. In contrast, the models estimated using the Poisson regressions pass the RESET test, i.e., the RESET test provides no evidence of misspecification of the gravity equations estimated using the PPML. With this particular specification, the model estimated using Tobit also passes the test for the traditional gravity equation.

Finally, we also check whether the particular pattern of heteroskedasticity assumed by the models is appropriate. As explained in section 3.2, the adequacy of the log-linear model was checked using the Park-type test, whereas the hypothesis $V[y_i|x] \propto \mu(x_i\beta)$ was tested by evaluating the significance of the coefficient on the term $\ln(\tilde{y}_i) \sqrt{\tilde{y}_i}$ in the Gauss-Newton regression indicated in expression (13). The p-values of the tests are reported in Table 4. Again, the log-linear specification is unequivocally rejected. On the other hand, these results indicate that the estimated coefficient on $\ln(\tilde{y}_i) \sqrt{\tilde{y}_i}$ is insignificantly different from zero at the usual 5 percent level. This implies that the Poisson PML assumption, $V[y_i|x] = \lambda_0 E[y_i|x]$ cannot be rejected at this significance level.

Table 4: Results of the tests for type of heteroskedasticity (p-values).

Test (Null hypothesis)	Trade > 0	Full sample
GNR ($V[y_i x] \propto \mu(x_i\beta)$)	0.144	0.115
Park (OLS is valid)	0.000	0.000

5.2.2 The Anderson-van Wincoop Gravity Equation

Table 5 presents the estimated coefficients for the Anderson-van Wincoop (2003) gravity equation, which controls more properly for multilateral resistance terms by introducing exporter and importer specific effects.

As before, the columns show, respectively, the estimated coefficients obtained using OLS on the log of trade, OLS on $\ln(1 + T_{ij})$, Tobit, NLS, PPML on the positive-trade

sample, and PPML. Note that, because this exercise uses cross sectional data, we can only identify bilateral variables.²⁸

As with the standard specification of the gravity equation, we find that using the Anderson-van Wincoop (2003) specification, estimates obtained with the Poisson method vary little when only the positive-trade subsample is used. Moreover, we find again strong evidence that the errors of the log-linear model estimated using the sample with positive-trade are heteroskedastic. With this specification, the two-degrees-of-freedom special case of White's test for heteroskedasticity leads to a test statistic of 469.2 and a p-value of zero.

Since we are now conditioning on a much larger set of controls, it is not surprising to find that most coefficients are sensitive to the introduction of fixed effects. For example, in the Poisson method, although the distance elasticity remains about the same and the coefficient on common colonial ties is still insignificant, the effect on common language is now smaller and the coefficient on free-trade agreements is larger. The results of the other estimations methods are generally much more sensitive to the inclusion of the fixed effects.

Comparing the results of PPML and OLS for the positive-trade subsample, the following observations are in order. The distance elasticity is substantially larger under OLS (-1.35 versus -0.75). Sharing a border has a positive effect on trade under Poisson, but no significant effect under OLS. Sharing a common language has similar effects under both techniques. Common colonial ties have strong effects under OLS (with an average enhancement effect of 100 percent), whereas Poisson predicts no significant effect. Finally, both techniques produce reasonably similar estimates for the coefficient on the trade-agreement dummy, implying a trade-enhancement effect of the order of 40 percent.

²⁸Since income elasticities cannot be separately identified, we use trade as the dependent variable of the gravity equation and not the ratio of trade to the product of GDP's, as in Anderson-van Wincoop (2003).

Like before, the other estimation methods lead to some puzzling results. For example, OLS on $\ln(1 + T_{ij})$ now yields a significantly negative effect of contiguity and under NLS, the coefficient on common-colonial ties becomes significantly negative.

Finally, we performed the same set of specification tests used before. The p-values of the heteroskedasticity-robust RESET test at the bottom of table 6 suggest that with the Anderson-van Wincoop (2003) specification of the gravity equation, only the models estimated by the PPLM method are adequate. The p-values of the tests to check whether the particular pattern of heteroskedasticity assumed by the models is appropriate are reported in Table 6. As in the traditional gravity equation, the log-linear specification is unequivocally rejected. On the other hand, these results indicate that the estimated coefficient on $\ln(\tilde{y}_i) \sqrt{\tilde{y}_i}$ is insignificantly different from zero at the usual 5 percent level. This implies that the Poisson PML assumption, $V[y_i|x] = \lambda_0 E[y_i|x]$ cannot be rejected at this significance level.

Table 5: The Anderson-van Wincoop Gravity Equation.

Estimator	OLS	OLS	Tobit	NLS	PPML	PPML
Dependent Variable	$\ln(T_{ij})$	$\ln(1 + T_{ij})$	$\ln(a + T_{ij})$	T_{ij}	$T_{ij} > 0$	T_{ij}
Log distance	-1.347** (0.031)	-1.332** (0.036)	-1.272** (0.029)	-0.582** (0.088)	-0.770** (0.042)	-0.750** (0.041)
Contiguity dummy	0.174 (0.130)	-0.399* (0.189)	-0.253 (0.135)	0.458** (0.121)	0.352** (0.090)	0.370** (0.091)
Common-language dummy	0.406** (0.068)	0.550** (0.066)	0.485** (0.057)	0.926** (0.116)	0.418** (0.094)	0.383** (0.093)
Colonial-tie dummy	0.666** (0.070)	0.693** (0.067)	0.650** (0.059)	-0.736** (0.178)	0.038 (0.134)	0.079 (0.134)
Free-trade agreement dummy	0.310** (0.098)	0.174 (0.138)	0.137** (0.098)	1.017** (0.170)	0.374** (0.076)	0.376** (0.077)
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	9613	18360	18360	18360	9613	18360
RESET test p-values	0.000	0.000	0.000	0.000	0.564	0.112

Table 6: Results of the tests for type of heteroskedasticity (p-values).

Test (Null hypothesis)	Trade > 0	Full sample
GNR ($V[y_i x] \propto \mu(x_i\beta)$)	0.100	0.070
Park (OLS is valid)	0.000	0.000

To sum up, whether or not fixed effects are used in the specification of the model, we find strong evidence that estimation methods based on the log-linearization of the gravity equation suffer from severe misspecification, which hinders the interpretation of the results. NLS is also generally unreliable. In contrast, the models estimated by PPML show no signs of misspecification and, in general, do not produce the puzzling results generated by the other methods.²⁹

6. Conclusions

In this paper, we argue that the standard empirical methods used to estimate gravity equations are inappropriate. The basic problem is that log-linearization (or, indeed, any non-linear transformation) of the empirical model in the presence of heteroskedasticity leads to inconsistent estimates. This is because the expected value of the logarithm of a random variable depends on higher-order moments of its distribution. Therefore, if the errors are heteroskedastic, the transformed errors will be generally correlated with the covariates. An additional problem of log-linearization is that it is incompatible with the existence of zeroes in trade data, which led to several unsatisfactory solutions, including truncation of the sample (that is, elimination of zero-trade pairs) and further non-linear transformations of the dependent variable.

²⁹It is worth noting that the large differences in estimates among the various methods persist when we look at a smaller subsample of countries that account for most of world trade and, quite likely, have better data. More specifically, we run similar regressions for the subsample of 63 countries included in Frankel (1997)'s study. These countries accounted for almost 90 percent of the world trade reported to the United Nations in 1992. One advantage of this subsample is that the number of zeroes is significantly reduced. Heteroskedasticity, however, is still a problem: The null hypothesis of homoskedasticity is rejected in both the traditional and the fixed-effects gravity equations. As before, PPML generates a smaller role for distance and common language than OLS, and, unlike OLS, PPML predicts no role for colonial ties. In line with the findings documented in Frankel (1997), the OLS estimated coefficient on the free-trade-agreement dummy is negative in both specifications of the gravity equation, whereas PPML predicts a positive and significant effect (slightly bigger than that found for the whole sample. This results are available – at request – from the authors.